Scattering Mechanism Inspired Non-Gaussian Diffusion Model for Polarimetric SAR Image Classification

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***Abstract*—The diffusion model has achieved excellent performance in natural image processing, which can learn the noise distribution through the degradation and restoration processes. However, the model is limited to Gaussian noise. Actually, polarimetric synthetic aperture radar (PolSAR) images have complex non-Gaussian speckle noises, for which the Gaussian diffusion model is difficult to learn their intrinsic statistical characteristics. In this article, we propose a novel scattering mechanism inspired non-Gaussian diffusion model for PolSAR image classification. To better simulate the PolSAR speckle noise, a mixed noise distribution is defined for PolSAR covariance matrices by combining Gamma multiplicative and Gaussian additive noises. A non-Gaussian forward noising process is derived to degrade a clean PolSAR image to a noisy image in steps. Then, the U-Net structure is trained to remove noise for each step, effectively extracting non-Gaussian statistical features. However, statistical features can only characterize the overall distribution of the dataset, which is insufficient to describe complicated individual objects; the original PolSAR data reflect the detailed scattering mechanism for individual pixels, which can provide complementary object information for classification. Therefore, a scattering–statistical joint learning network is further developed with a dual-branch architecture to enhance the discrimination ability. In particular, a multiscale pyramid module and attention mechanism are designed to improve the ability of feature learning. Experimental results on five real PolSAR datasets demonstrate that the proposed method effectively captures edge details and preserves homogeneous regions for terrain classification, especially in heterogeneous regions.**

***Index Terms*—Dual-branch network, non-Gaussian diffusion model, polarimetric synthetic aperture radar (PolSAR) image classification, scattering–statistical joint learning network.**

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I. Introduction

THE polarimetric synthetic aperture radar (PolSAR) system can emit and receive electromagnetic waves with four polarimetric modes, formulating a 2×2 scattering matrix for each resolution unit [1]. PolSAR system can work on any weather condition and in all day and night, which can be applied to military and civilian tasks [2], [3], [4]. With rich scattering information, PolSAR images have found wide use in terrain classification [5], [6], object recognition [7], change detection [8], and disaster prediction [9], [10]. Among these applications, land cover classification stands out as a crucial task that has garnered significant interest from the research community.

Researchers have explored a variety of traditional classification techniques, including those based on statistical distributions [11], [12], scattering mechanisms [13], [14], and machine learning [15]. The goal of these methods is to effectively leverage the polarimetric data to achieve high-precision classification results. However, in heterogeneous areas, the significant scattering variations within identical terrain objects pose a challenge for accurate classification without the incorporation of semantic features.

Recently, deep learning has achieved excellent performance in natural image processing and has rapidly expanded to the PolSAR image classification task. Mainstream methods include convolutional neural network (CNN), graph convolutional network (GCN), and Transformer [16], [17], [18]. All of them can achieve effective applications for PolSAR image classification. Among them, CNN-based methods are well-suited for image processing tasks, as they excel at capturing local spatial features through the convolutional operation. Considering polarimetric characteristics, some variants are developed, including scattering information-based CNN [19], complex-valued CNN [20], graph convolution combined with CNN [21], and others. For example, Zhang et al. [22] proposed a CNN model to jointly learn scattering similarity and textural features. Alkhatib et al. [23] proposed an attention-based

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complex-valued CNN method to enhance classification performance. Shi et al. [21] proposed a CNN-improved fuzzy GCN to learn both local and global features for PolSAR image classification. These methods can learn polarimetric information effectively by involving various scattering features, while some global information may be ignored. Recently, Transformers are popular due to their global learning capability with the multihead attention mechanism. Jamali et al. [17] extended the Transformer method to PolSAR images by proposing a local window attention transformer. Xin et al. [24] considered a dual-frequency PolSAR image and constructed a transformer model based on mixed patch interaction. However, existing deep learning models generally work with spatial convolution, ignoring the statistical features of the PolSAR data. Each PolSAR data are listed as a vector, and convolution in each layer can be considered a nonlinear operations that cannot describe the statistical distribution of PolSAR data effectively.

Recent diffusion models [25], [26], as effective generative methods, can effectively learn statistical features by adding Gaussian noise and have been successful in the field of natural image generation. Their advantages have also been extended to remote sensing image processing tasks [27], [28], [29], such as super-resolution and classification of hyperspectral images. For example, Chen et al. [30] proposed the SpectralDiff model for hyperspectral image classification, which explored the distribution information of high-dimensional, high-redundancy data through iterative denoising and generation processes. Furthermore, Zhou et al. [31] explored a multitime step and multistage diffusion feature learning framework, addressing the limitations of manual feature selection. Zhang et al. [32] used the distribution knowledge learned by diffusion models as supplementary information to the original data for contrastive learning. Yang et al. [33] incorporated the manifold structures between samples into the objective constraints and combined adversarial networks to capture the true manifold structures in the data distribution, and so forth. Nevertheless, these methods are limited to the Gaussian noise distribution. Some researchers have applied diffusion models to SAR image denoising. For instance, Guha and Acton [34] derived mathematical formulas for the forward process, the reverse process, and training objectives to achieve denoising of speckle multiplicative noise in SAR images. Perera et al. [35] added speckle noise to blur clean images and used conditional diffusion models, with time steps and clean images as conditions, to denoise SAR images. Zhou et al. [36] introduced diffusion models into SAR target detection. These advances highlight the growing versatility and effectiveness of diffusion models across various image processing tasks.

However, PolSAR images exhibit distinct characteristics compared with both optical and hyperspectral images due to their intricate scattering properties and speckle noise [37]. Each pixel in PolSAR data is represented by a covariance matrix, in which diagonal elements are single-polarimetric SAR for each channel and adhere to a multiplicative Gamma distribution. On the other hand, nondiagonal elements represent the channel-correlation information, manifesting as a mixture of multiplicative Gamma and additive Gaussian distributions. Therefore, PolSAR speckle is a type of multiplicative

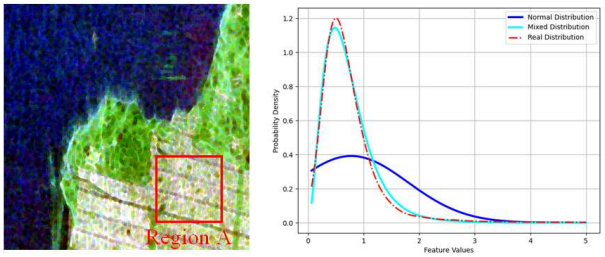


Fig. 1. Distribution of PolSAR covariance matrix on San Francisco datasets. (a) Pauli RGB image. (b) Different distributions to fit the real distribution of PolSAR covariance matrices.

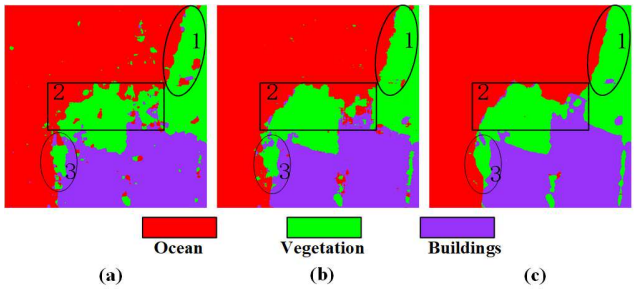


Fig. 2. Classification maps of different distribution models on San Francisco dataset. (a) Gaussian diffusion model. (b) Proposed non-Gaussian diffusion model. (c) Proposed scattering-statistical joint learning method.

and additive mixed noise [37], [38]. Current diffusion models fail to model PolSAR data distribution effectively. Up to date, there is no related complicated non-Gaussian distribution-based diffusion model for PolSAR image classification. Existing diffusion methods are based on the additional Gaussian distribution noise. Incorporating non-Gaussian speckle noise characteristics into the diffusion process better reflects the true statistical nature of PolSAR data, thereby allowing the model to capture intrinsic features that are more representative of PolSAR imagery. Therefore, it is essential to develop a novel non-Gaussian diffusion model specifically tailored for PolSAR image processing.

To further analyze this issue, we take the San Francisco PolSAR dataset as an example in Fig. 1(a). Region A is sampled to test the distribution of the PolSAR covariance matrix. As shown in Fig. 1(b), the red line is the real-data distribution. The cyan line is fit by the mixed distribution, while the blue line is fit by the Gaussian distribution. It can be seen that the data distribution is better fit by the proposed mixed distribution rather than the Gaussian distribution.

Furthermore, to verify the effect of different distribution-based diffusion models on classification, a Gaussian-based diffusion method [30] (noted by "Gaussian") and our proposed non-Gaussian mixed diffusion method (noted by "non-Gaussian") are conducted on Fig. 1(a), respectively. Experimental results are shown in Fig. 2(a) and (b). It can be seen from (a) that the Gaussian method appears to have many noisy classes in Ocean and Vegetation areas, since the Gaussian distribution cannot model complicated PolSAR data,

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especially the PolSAR speckle. The proposed non-Gaussian method in (b) can obtain a more homogeneous classification map, with less misclassification in Ocean and Vegetation classes. It is because the non-Gaussian model can better model the distribution of PolSAR speckle data and obtain better statistical features. Therefore, a novel non-Gaussian diffusion model is necessary to be investigated for PolSAR image classification.

In addition, from Regions 1 to 3 in Fig. 2(a) and (b), it can be seen that there still exist noisy classes for both methods, since only statistical information is not enough to describe individual features of complicated terrain objects. The original PolSAR data contain rich individual scattering features, which can provide complementary information for classification. In Fig. 2(c), we further develop a dual-branch network to combine the advantages of both non-Gaussian diffusion and scattering information to enhance classification.

Based on the analysis above, we propose a novel scattering mechanism inspired non-Gaussian diffusion model to effectively learn the complicated distribution of the PolSAR covariance matrix for the classification of PolSAR images. First, a Gamma and Gaussian mixed model is designed to simulate the multiplicative and additive mixed noises. This mixed noises are added in the diffusion model, which allows the model to capture and learn the intrinsic distribution characteristics of polarimetric SAR images. To learn the reverse process for each time step, a denoising U-Net network is utilized to learn the statistical features. To combine complementary features, a dual-branch network is designed to further jointly learn statistical and scattering features for refined classification. A new loss function is defined by considering the cross-entropy loss, as well as both the intraclass and interclass losses. The main contribution can be concluded as follows.

1) To the best of the authors' knowledge, this is the first time to propose a novel non-Gaussian diffusion model for PolSAR image classification. We model the covariance matrix as a multiplicative and additive mixed distribution, and infer the forward diffusion procedure with non-Gaussian distribution. It overcomes the Gaussian limitation and better describes the statistical features of PolSAR speckle noises.
2) To jointly learn scattering and statistical features, a dual-branch attention network is designed to combine statistical features from the non-Gaussian diffusion model and scattering mechanism-based features from the original data together, which can make full use of both statistical and individual information to enhance classification.
3) A cosine similarity loss and an interclass discriminative loss are designed for dual-branch networks to increase both the intraclass similarity and interclass discriminating capability. The mixed loss function can enhance the discriminating ability of the proposed network.

The rest of this article is organized as follows. Section II is the background. The proposed method is introduced in Section III in detail. Experimental results and analysis are exhibited in Section IV. Section V provides the conclusion.

II. Background

A. Diffusion Model

Diffusion models are the popular methods for image generation, which have achieved outstanding performance due to their diffusion procedure. The fundamental principle involves the construction of a Markov chain that progressively introduces Gaussian noise, simulating the image degradation process [25], [30]. This method can learn the noisy distribution introduced at each step of this process, which aims to recover the image from the noisy state back to the clear state. It is performed with a forward process by gradually adding Gaussian noise by t times from the original image to the noisy image. Then, a training process is conducted to learn the noisy distribution in each step.

For the forward process, the noise is randomly sampled from a Gaussian distribution $N(0, I)$. Then, the noisy image in tth step is obtained by adding noise t times iteratively from the original image x_0. It can be expressed as $x_t = (\bar{a}_t)^{1/2}x_0 + (1 - \bar{a}_t)^{1/2}\varepsilon_t$ and $\bar{\alpha}_t = \prod_{s=0}^{t} a_s$, where α is a hyperparameter that regulates the level of noise addition and ε is the sampling noise from the Gaussian distribution. In contrast, during the reverse process, the generated image is reconstructed as a denoising procedure by step-by-step noise prediction. The loss function of the diffusion model is defined as follows:

$$\mathcal{L} = E_{t\sim[1:T],x_0,\varepsilon_t}\left[\left\|\varepsilon_t - \varepsilon_\theta\left(\sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\varepsilon_t, t\right)\right\|^2\right] \quad (1)$$

where ε_t represents the actual noise introduced at time step t, $\varepsilon_\theta(\cdot,\cdot)$ is the noise predicted by the model, and $\|\cdot\|$ represents the $L2$-norm. $(\bar{\alpha}_t)^{1/2}x_0 + (1 - \bar{\alpha}_t)^{1/2}\varepsilon_t$ is the noisy image in time step t. T indicate the total number of time steps.

Algorithm 1 Improved Sampling Algorithm [39]

Input: A degraded sample x_t

for $t = T$ **down to** 1 **do**
  $\hat{x}_0 = \theta(x_t, t)$
  $x_{t-1} = x_t - D(\hat{x}_0, t) + D(\hat{x}_0, t-1)$
end for

For non-Gaussian data, it cannot learn the reverse process like DDPM. Fortunately, cold diffusion [39] is proposed, which extends the image degradation mode of diffusion to arbitrary forms of noise. The reverse process is a reconstructed process by a U-Net, which is not related to the adding noise model. According to the cold diffusion, an iterative improved sampling algorithm in Algorithm 1 is proposed. It can be highly effective in recovering high-quality results. The loss function is expressed as follows:

$$\mathcal{L} = \mathbb{E}_{x_0}\left\|\theta\left(D\left(x_0, t\right), t\right) - x_0\right\| \quad (2)$$

where $D(x_0, t)$ represents the degradation operation from the original image x_0 to the noisy image in time step t and $\theta(\cdot,\cdot)$ is a restoration function from noisy to clean images.

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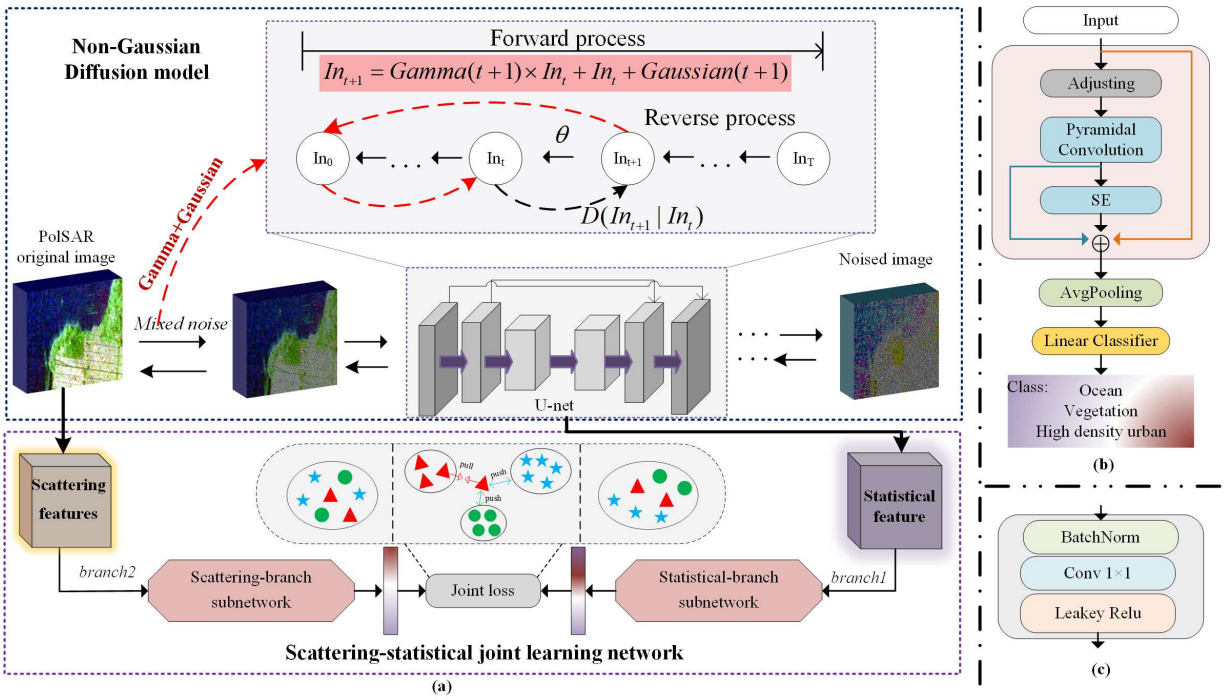


Fig. 3. Overview architecture of the proposed scattering mechanism inspired non-Gaussian diffusion model for polarimetric SAR image classification. (a) Framework of the proposed NGDiff_SM method. (b) Network structure of the statistical-/scattering-branch subnetwork. (c) Adjusting module.

B. PolSAR Data Distribution

For PolSAR images, each resolution unit is a 2×2 scattering matrix extracted from four polarimetric modes, expressed as follows:

$$\mathbf{S} = \begin{bmatrix} S_{\mathrm{hh}} & S_{\mathrm{hv}} \\ S_{\mathrm{vh}} & S_{\mathrm{vv}} \end{bmatrix} \tag{3}$$

where hh represents the electromagnetic wave from horizontal emitting and horizontal receiving. hv is the wave from horizontal emitting and vertical receiving. Under the Pauli base condition, $S_{\mathrm{hv}} = S_{\mathrm{vh}}$. Then, the scattering matrix can be vectored as $k = [S_{\mathrm{hh}}, (2)^{1/2} S_{\mathrm{hv}}, S_{\mathrm{vv}}]$. To better reflect the scattering characteristics of terrain objects, a covariance matrix, most commonly used PolSAR data format, can be derived by

$$\mathbf{C} = k \times k^H = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}. \tag{4}$$

It is well known that the diagonal elements are real numbers, while the nondiagonal ones are complex. As to their distribution, for the diagonal elements in the covariance matrix $\mathbf{C}$, they can be modeled as the multiplicative noises, while nondiagonal elements are mixed distribution with multiplicative and additive models [37]. Therefore, covariance matrices can be modeled as a complicated non-Gaussian distribution.

III. Proposed Method

In this article, a novel scattering mechanism inspired non-Gaussian diffusion model(NGDiff_SM) is developed for PolSAR image classification, of which the framework is illustrated in Fig. 3. It mainly consists of a novel non-Gaussian diffusion model and a scattering–statistical joint learning network. First, a non-Gaussian diffusion model is proposed for PolSAR image classification, which utilizes the Gamma and Gaussian mixed distribution to better model the complicated PolSAR data. In the forward process, an adding noise model is inferred with the defined non-Gaussian mixed distribution for t time steps. During the reverse process, the U-Net is utilized to learn the non-Gaussian noise at each step, thereby achieving an in-depth understanding of the statistical features of polarimetric SAR images. Second, a scattering–statistical joint learning network is designed to enhance classification performance. Scattering information from original data and statistical features from a non-Gaussian diffusion model are input into a dual-branch network to enhance classification. Finally, the interclass, intraclass, and cross-entropy losses are defined to enhance the discriminating ability of the network.

A. Non-Gaussian Diffusion Model

1) Forward Non-Gaussian Diffusion Process: Given a polarimetric SAR image I, mixed noise, comprising both multiplicative and additive components, is added to the polarimetric SAR data I. Assuming that a multiplicative noise m_i is added to image I, expressed as $y = m_i I$. In addition, for the additive noise n_i, the original image can be added as $y = I + n_i$. Therefore, for non-Gaussian with the additive and multiplicative mixed noise, the noisy image is defined as $y = m_i I + I + n_i$.

In this article, for the diagonal elements in the covariance matrix, we define the multiplicative noise as the random Gamma noise, i.e., $m_i = \mathrm{Gamma}(k, \theta)$, where k and θ are the shape and scale parameters. For nondiagonal elements, the

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mixed speckle noises with Gamma and Gaussian distributions are defined. The Gamma noise is randomly selected from m_i. The Gaussian noises are given as $n_i = \alpha_i \cdot \varepsilon$, $\alpha_i \in (0.0001, 0.02)$, ε denotes a random Gaussian noise with zero mean and unit variance, i.e., $\varepsilon \sim \mathcal{N}(0, 1)$.

Based on the principles of Markov chains, the data distribution at time step $t+1$ depends solely on the data distribution at time step t. In the forward process, each image block is extracted as the clean input instance I_0. Mixed noises are randomly generated to add to different time steps t. When $t = 0$, the image is I_0. When $t = 1$, $I_1 = m_1 \cdot I_0 + I_0 + n_1$. With the time step increasing, $I_{t+1} = m_{t+1} \cdot I_t + I_t + n_{t+1}$, where $m_{t+1} \in \mathbb{R}^{n \times n \times D}$ and $n_{t+1} \in \mathbb{R}^{n \times n \times D}, t \in (0, T)$ are the Gamma and Gaussian noises, respectively. T is the maximum time step. Based on the analyze above, the noisy image I_t at time step t is only related to I_0, derived by

$$I_t = \prod_{i=1}^{t} (m_i + 1) \cdot I_0 + \sum_{i=1}^{t} n_i \prod_{j=i+1}^{t} \left(m_j + 1\right). \tag{5}$$

That is,

$$I_t = \alpha_t \cdot I_0 + \beta_t \tag{6}$$

where $\alpha_t = \prod_{i=1}^{t}(m_i + 1)$ and $\beta_t = \sum_{i=1}^{t} n_i \prod_{j=i+1}^{t}(m_j + 1)$. In (6), given the Gamma noises m_i and Gaussian noises n_i,α_t and β_t can be derived directly. Thus, the noised image I_t in step t can be calculated. During the calculation of α_t and β_t, two lists are defined to reduce repeated calculation. One list is designed to store the random Gamma noise for each time step, defined as $M = \{m_1, m_2, \ldots, m_T\}$. Another list M_{reverse} is defined to store the accumulative multiplication in β_t, noted by

$$M_{\text{reverse}}^{t} = \left\{ \prod_{1}^{t} (m_i + 1), \ldots, \prod_{t-1}^{t} (m_i + 1), (m_t + 1) \right\}. \tag{7}$$

Thus, the forward process can be calculated effectively.

2) Reverse Process With U-Net Network: Since the speckle noise follows a non-Gaussian distribution, the reverse process of DDPM based on Gaussian assumptions is no longer suitable. Based on the cold diffusion model, in this article, we adopt a U-Net network combined with the improved sampling algorithm (see Algorithm 1) to directly learn the reverse process, recovering noisy images to clean ones for arbitrary distribution. This approach enables the diffusion network to effectively adapt to non-Gaussian modeling through the U-Net.

3) Network Training: For PolSAR instance $I_0 \in R^{n \times n \times D}$, for each time step t, we generate multiplicative and additive noise m_t and n_t. Then, α_t and β_t are calculated. According to the forward process, the mixed noise I_t is achieved. According to the time step t and the noisy image I_t, the U-Net is trained to generate the obscuring image. The loss function is defined as follows:

$$\mathcal{L} = E_{I_n \sim I} \left[\| \theta(I_t, t) - I_0 \| \right] \tag{8}$$

where $\theta(\cdot, \cdot)$ represents the U-Net. $\| \cdot \|$ is $L2$-norm. This loss function minimizes the distance between the predicted image and the original clean image. According to the loss function, the U-Net is trained for each time step. Thus, the learned features for each step can be extracted, which can effectively reflect the speckle noisy distribution for PolSAR images.

B. Scattering and Statistical Joint Dual-Branch Network

To effectively model PolSAR data, statistical features can be extracted at each time step t, using a non-Gaussian diffusion process, which gradually introduces non-Gaussian noise. These features, denoted by I_{feature}, are obtained from a reversed U-Net, enabling to model the statistical distribution of covariance matrices. However, relying solely on statistical features may not fully capture the refined discriminative characteristics, as they lack individual information. Scattering features, which are well-suited for representing target information, have been widely employed in the PolSAR classification task. To leverage the advantages of both feature types, a dual-branch network is proposed, enabling the joint learning of statistical and scattering features for enhanced classification performance. The whole network structure is illustrated in Fig. 4. First, statistical features are extracted from the non-Gaussian diffusion model, while scattering features are derived from the original data, serving as separate inputs to the network. The dual-branch network design enhances the learning of both feature types independently. Each branch incorporates a pyramid module to capture multiscale features and an squeeze-and-excitation (SE) module to optimize the channelwise information. Compared with multiscale convolutional networks such as inception modules [40] and dilated convolutions [41], the pyramid network constructs a hierarchical architecture. This design facilitates multiscale feature fusion by combining low-level spatial details with high-level semantic representations, thereby enhancing the network's capacity to capture both fine-grained and semantic information across different layers. In addition, by incorporating the SE module, the network can effectively model the interdependencies between channels, enabling it to emphasize informative features while suppressing useless ones, thus improving representational capacity with the minimal computational cost. Finally, scattering and statistical features are fused for joint learning and classification.

The network takes two types of inputs: the PolSAR scattering feature $x \in R^{H \times W \times D_1}$ and statistical feature $f \in R^{H \times W \times D_2}$, extracted from the proposed non-Gaussian diffusion model. Here, H and W are the size of the PolSAR image. D_1 and D_2 are the channel dimensions of scattering and statistical features, respectively. Generally, D_2 is relatively high due to the diffusion learning process. To reduce computational complexity, principal component analysis (PCA) is applied to f, reducing its dimensionality while preserving its main components, resulting in a feature $g \in R^{H \times W \times D_3}$. The PolSAR scattering feature is derived directly from the original data, which has been widely used as input in deep learning-based PolSAR image classification. Here, the PolSAR covariance matrix is converted into the 9-D vector, defined as follows:

$$V = \big[C_{11}, C_{22}, C_{33}, \text{real}\,(C_{12}), \text{img}\,(C_{12}), \text{real}\,(C_{13}) \\ \text{img}\,(C_{13}), \text{real}\,(C_{23}), \text{img}\,(C_{23})\big] \tag{9}$$

where real$(\cdot)$ denotes the real part of a complex number, while img$(\cdot)$ represents its imaginary part.

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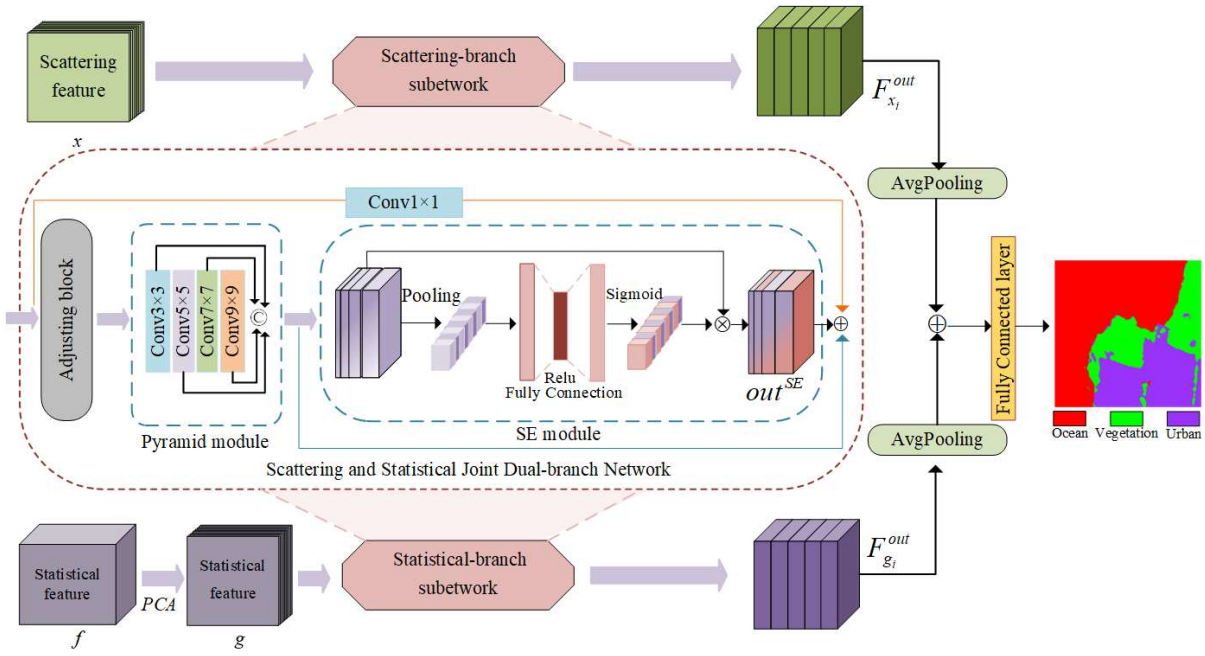


Fig. 4. Framework of the scattering–statistical joint learning network for PolSAR image classification.

The dual-branch subnetwork consists of an adjusting block, pyramid module, and SE module. The adjusting block includes batch normalization, a 1×1 convolution, and a LeakyReLU layer for data normalization and dimensional adjustment. The pyramid module extracts multiscale features using four convolutional kernels: 3×3, 5×5, 7×7, and 9×9, respectively. The output is the contacting features from all four scales, defined as follows:

$$\begin{aligned} \text{out}_{x_i}^{\text{pyramid}} = \text{Concat}\left\{\text{Conv}^{3\times3}\left(x_i\right), \text{Conv}^{5\times5}\left(x_i\right)\right. \\ \left.\text{Conv}^{7\times7}\left(x_i\right), \text{Conv}^{9\times9}\left(x_i\right)\right\} \end{aligned} \tag{10}$$

where Concat$(\cdot)$ is the concatenation operation. $\text{out}_{x_i}^{\text{pyramid}}$ is the output of the pyramid module.

In addition, an SE module is incorporated to enhance the discriminative capability of the channel features. The SE module is defined as follows:

$$\begin{aligned} \text{out}_{x_i}^{\text{SE}} &= \text{SE}\left(\text{out}_{x_i}^{\text{pyramid}}\right) \\ &= \text{out}_{x_i}^{\text{pyramid}} \otimes f_\text{sig}\left(f_\text{relu}\left(\text{Pooling}\left(\text{out}_{x_i}^{\text{pyramid}}\right)\right)\right) \end{aligned} \tag{11}$$

where $\text{out}_{x_i}^{\text{SE}}$ is the output of SE module. $f_\text{sig}(\cdot)$ represents the dimensionality expansion operation using fully connected layers and the sigmoid function. $f_\text{relu}(\cdot)$ is the dimensionality reduction operation using fully connected layers and the ReLU function. Pooling$(\cdot)$ is the global pooling, and $\otimes$ denotes the pixelwise multiplication. In addition, residual connections from different layers are incorporated to enhance the feature representation and capture complementary information. It is expressed as follows:

$$F_{x_i}^{\text{out}} = \text{Relu}\left(\text{out}_{x_i}^{\text{SE}} \oplus \text{out}_{x_i}^{\text{pyramid}} \oplus \text{Conv}^{1\times1}\left(x_i\right)\right) \tag{12}$$

where $\oplus$ is the pixelwise connection. $\text{Conv}^{1\times1}(\cdot)$ is a 1×1 convolution to adjust feature dimension. $F_{x_i}^{\text{out}}$ is the learned feature from the proposed subnetwork. After feature extraction from two branch networks, the statistical and scattering features, denoted by F_1 and F_2. These features are then combined for classification. The entire feature learning process of the two-branch network can be expressed as follows:

$$C = \text{softmax}\left(\text{FC}\left(\text{branch1}\left(\text{NGD}\left(I, t\right)\right) \oplus \text{branch2}\left(V\right)\right)\right) \tag{13}$$

where I is the input clean image. $\text{NGD}(I, t)$ represents feature extraction from the non-Gaussian diffusion model in time step t. V is the original PolSAR scattering vector from covariance matrix. FC$(\cdot)$ is the fully connected layer.

C. Loss Function

The dual-branch network learns statistical and scattering features separately, enhancing the diversity of feature representation for each pixel. However, the representations from the two subnetworks for each pixel should be similar, ensuring they are classified into the same class. This is akin to a multimodal feature learning process, where the intraclass similarity for the same sample should be maximized. Moreover, for two pixels i and j with different classes, their features should be distinct, ensuring that they are assigned to different classes. Therefore, the interclass diversity should be maximized. To improve the class discriminability, we define the intraclass similarity and interclass variability.

Therefore, the final loss function consists of cross entropy, intraclass, and interclass loss, expressed as follows:

$$\begin{aligned} \mathcal{L} = \mathcal{L}\left(\mathbf{y}, \mathbf{p}\right) + \mathcal{L}\left(F_{x_i}^{\text{out}}, S_{x_i}^{\text{out}}\right) \\ + \mathcal{L}_{\text{inter}}\left(F_{x_i}^{\text{out}}, F_{x_j}^{\text{out}}\right) + \mathcal{L}_{\text{inter}}\left(S_{x_i}^{\text{out}}, S_{x_j}^{\text{out}}\right) \end{aligned} \tag{14}$$

where $F_{x_i}^{\text{out}}$ and $S_{x_i}^{\text{out}}$ are the feature output from scattering and statistical subnetworks, respectively. $\mathcal{L}(\mathbf{y}, \mathbf{p})$ is the cross-entropy loss, $\mathcal{L}(F_{x_i}^{\text{out}}, S_{x_i}^{\text{out}})$ is the intraclass similarity from the

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same sample, and $\mathcal{L}_{\text{inter}}(F_{x_i}^{\text{out}}, F_{x_j}^{\text{out}})$ is the interclass variation in scattering subnetwork from different samples. $\mathcal{L}_{\text{inter}}(S_{x_i}^{\text{out}}, S_{x_j}^{\text{out}})$ is the interclass variation in statistical subnetwork. The cross-entropy loss is defined as follows:

$$\mathcal{L}(\mathbf{y}, \mathbf{p}) = -\sum_{i=1}^{C} y_i \log(p_i) \tag{15}$$

where y_i is the real label vector. p_i is the obtained class probability after the softmax layer, which can be expressed by

$$\mathbf{p} = \text{Softmax}\left(\text{FC}\left(\text{Pooling}\left(F_{\text{out}}\right)\right)\right). \tag{16}$$

In the multimodal representation of PolSAR data, two subnetworks can map the two modal features into another subspace after feature learning. Two types of representations from the same sample should have high similarity, while different samples are far from each other. The feature similarity can be effectively measured by the cosine similarity, defined as follows:

$$\mathcal{L}(c) = 1 - \frac{F_{x_i}^{\text{out}} \cdot S_{x_i}^{\text{out}}}{\left|F_{x_i}^{\text{out}}\right| \times \left|S_{x_i}^{\text{out}}\right|} \tag{17}$$

where $F_{x_i}^{\text{out}}$ and $S_{x_i}^{\text{out}}$ are the output features from scattering and statistical subnetworks.

In addition, for different samples, the interclass variation is defined as follows:

$$\mathcal{L}_{\text{inter}}(i, j) = \frac{1}{C(C-1)} \sum_{i \neq j} d_{ij}^2 \tag{18}$$

where C_c is the sample number of class c. C is the class number. d_{ij} is the distance of class i and j, defined as follows:

$$d_{ij} = \left\| m_i - m_j \right\|_2 \tag{19}$$

m_c is the mean feature of class c, defined as follows:

$$m_c = \frac{1}{|C_c|} \sum_{f_n \in C_c} f_n \tag{20}$$

where f_n is the feature vector of sample n.

Algorithm 2 Scattering Mechanism Inspired Non-Gaussian Diffusion Model for Classification

Input: Original data **I**.

1. Construct non-Gaussian noise consisting of a mixture of Gamma and Gaussian distributions.
2. Add non-Gaussian noise to **I** (as in (5)) to obtain the degraded image $\mathbf{I_t}$.
3. Train the non-Gaussian diffusion U-net network, using the loss function defined in (8).
4. Extract the statistical features **f** of PolSAR data learned by the non-Gaussian diffusion network.**Classification:**
5. Obtain **x** from **I**; process **f** with PCA to obtain **g**.
6. Input **g** and **x** into the Scattering and Statistical Joint Dual-branch Network to train and obtain $F_{x_i}^{out}$ and $F_{f_i}^{out}$.
7. Calculate the $F_{x_i}^{out}$ and $F_{f_i}^{out}$ joint loss using (14).
8. Update the network by minimizing the joint loss.

Output: Predicted class label map **C**.

The whole algorithm procedure can be illustrated in Algorithm 2.

IV. Experimental Results and Analysis

A. Experimental Data and Settings

1) Experimental Data: To validate the effectiveness of the aforementioned method, a series of experiments were carried out on five PolSAR datasets, characterized by diverse bands and sensors. These datasets are extensively used in the field of PolSAR image classification. Their corresponding label maps are supplied for benchmarking purposes, as shown in Fig. 5. The black area on the label map indicates that the pixels are unlabeled. The details of the five datasets in this study are subsequently introduced as follows.

2) Xi'an Dataset: This image is C-band data from Xi'an, China, acquired by the RADARSAT-2 system. It consists of 512 × 512 pixels with a resolution of 8 × 8 m. The land cover mainly consists of three types: water, grass, and building. Fig. 5(a) shows the PauliRGB false-color image and its corresponding label map, respectively.

3) Flevoland1 Dataset: This dataset consists of C-band PolSAR data acquired over Flevoland, the Netherlands, spanning 1400 × 1200 pixels. The image contains four types of land covers: water, urban, woodland, and cropland. A corresponding false-color image with its label map is depicted in Fig. 5(b).

4) Oberpfaffenhofen Dataset: This dataset comprises the complete PolSAR L-band data captured by the ESAR system from Oberpfaffenhofen, Germany. Fig. 5(c) presents the PauliRGB image along with its corresponding label map. The image has the size of 1300 × 1200 pixels. Specifically, the left side presents the PauliRGB image, and the right side displays the label map, which encompasses five terrain types: bare ground, forest, building, farmland, and road.

5) San Francisco Dataset: The NASA/JPL AIRSAR system captured an L-band polarimetric SAR dataset over the San Francisco area in 1985, offering a detailed view of diverse landscapes. The dataset boasts a resolution of 900×1024 pixels and identifies five primary radar targets: bare soil, mountains, oceans, urban, and vegetation. Fig. 5(d) exhibits the PauliRGB image and label map for this dataset.

6) Flevoland2 Dataset: This dataset is the full polarimetric L-band data captured by the NASA/Jet AIRSAR system over Flevoland, an agricultural area of The Netherlands. The image has the size of 750 × 1024 pixels. Fig. 5(e) shows the PauliRGB image and the label map for this dataset. Specifically, the dataset encompasses 15 distinct classes, including stem beans, peas, forest, lucerne, beets, potatoes, bare soil, grasses, rapeseed, barley, water, buildings, and three types of wheat.

a) Comparative Algorithms: In addition, to assess the validity of our proposed methodology, we choose six benchmark algorithms as compared methods known for their outstanding performance in PolSAR image classification, including Super_RF [42], DFGCN [43], CVS3ANET [44], HybridCVNet [45], AMS-M2ESL [46], and SpectralDiff [30]. The details of comparative approaches are given as follows.

1) *Super_RF:* It utilizes statistical texture and polarization features to construct superpixels through a simple linear

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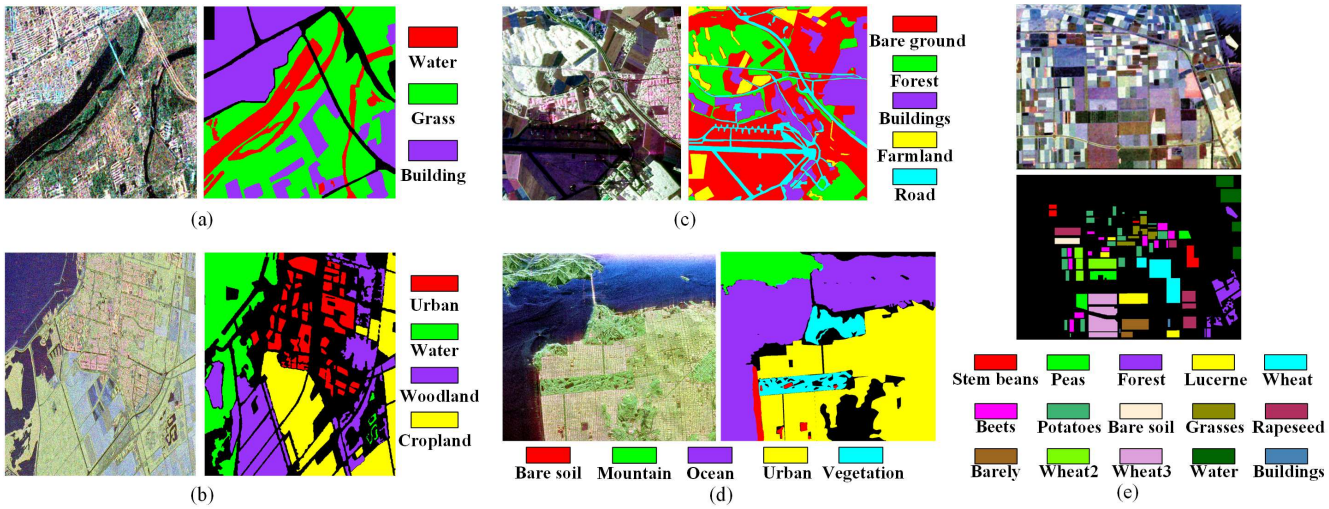


Fig. 5. Five datasets with their PauliRGB images and corresponding label maps. (a) Xi'an dataset. (b) Flevoland1 dataset. (c) Oberpfaffenhofen dataset. (d) San Francisco dataset. (e) Flevoland2 dataset.

TABLE I
CLASSIFICATION ACCURACY OF EACH COMPARISON ALGORITHM ON XI'AN DATASET(%)

class	Super_RF	DFGCN	CVS3ANET	HybridCVNet	AMS-M2ESL	SpectralDiff	proposed
Water	70.91	84.64	84.39	92.01	88.99	89.70	**92.02**
Grass	94.97	91.79	93.99	94.99	90.35	97.09	**97.43**
Building	90.94	87.21	96.72	95.05	90.27	97.74	**98.47**
OA	89.94	89.10	93.50	94.56	90.12	96.21	**96.99**
AA	85.61	87.88	91.70	94.02	89.87	94.85	**95.98**
KAPPA	83.02	81.84	89.24	91.02	83.68	93.73	**95.02**

iterative clustering algorithm, and then combines with the random forest algorithm to incorporate polarization features and spatial information together for improving classification accuracy.

2) *DFGCN:* It is a deep fuzzy GCN that employs fuzzy logic to construct adaptive fuzzy graphs, enhancing the accuracy of terrain classification by incorporating physical attributes and spatial information for both labeled and unlabeled pixels in homogeneous and heterogeneous landscapes.
3) *CVS3ANET:* This method gives a complex-valued spatial–scattering separated attention network for PolSAR image classification. It utilizes CV-3D convolutions to explore the features in both spatial and scattering dimensions, and reduces parameters via sequential processing. Furthermore, it incorporates SE attention mechanisms to enhance feature representation capabilities.
4) *HybridCVNet:* This method integrates CV-CNN with CV-ViT and achieves enhanced PolSAR classification by leveraging the complex-valued information of scattering targets along with the complementary information from various feature maps.
5) *AMS-M2ESL:* It introduces an adaptive mask sampling model that can capture more accurate features of ground objects. In addition, it gives a manifold-to-Euclidean subspace learning algorithm framework, which can better handle nonlinear features in hyperspectral images and extract more differentiated features.
6) *SpectralDiff:* This method presents an unsupervised generative framework for hyperspectral images, which defines a Gaussian-based diffusion module to learn statistical features, followed by an attention-based classification module to achieve pixel-by-pixel classification. It can learn spectral–spatial distributions of high-dimensional data and suppress redundant data and noisy information. In this experimental setup, the epoch is set to 2500 and extract Gaussian features with time step $t = 5$ for classification.

Comparative methods [items (1) –(5)] are carefully selected according to their demonstrated success and relevance in the field of PolSAR image classification, ensuring a rigorous evaluation of our proposed approaches. Both algorithm (6) and the proposed method employ diffusion models for classification, making them more compelling for comparative experimental analysis.

b) Experimental Settings: During the training of our non-Gaussian diffusion module, we configure several key parameters to optimize performance. Specifically, the patch size is set to 64, and the learning rate is established at 0.0001. The total time step T is set to 16, with a total of 500 epochs for the training process. The Gaussian distribution used in the noise term is parameterized with zero mean and unit variance, i.e., $\varepsilon \sim N(0, 1)$. The parameters of the Gamma distribution are estimated from the PolSAR image by the matrix log-cumulants (MoMLC) method [47], [48].

For the classification module, we adjust the patch size to 13, allocate 5% of the data to the training set, and 1% to the validation set. The proportion of training datasets used in the comparative experiments aligns with the proportions

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TABLE II

CLASSIFICATION ACCURACY OF EACH COMPARISON ALGORITHM ON THE FLEVOLAND1 DATASET(%)

class	Super_RF	DFGCN	CVS3ANET	HybridCVNet	AMS-M2ESL	SpectralDiff	proposed
Urban	81.84	91.83	**99.91**	97.00	84.80	92.76	99.66
Water	98.69	99.57	98.85	99.78	99.19	98.05	**99.89**
Woodland	94.92	95.93	97.11	98.31	93.40	96.82	**99.68**
Cropland	94.16	93.82	99.17	97.17	93.85	96.84	**99.79**
OA	93.88	95.69	98.61	98.17	93.91	96.59	**99.77**
AA	92.40	95.29	98.76	98.07	92.81	96.12	**99.75**
KAPPA	91.61	94.11	98.11	97.51	91.66	95.35	**99.68**

TABLE III

CLASSIFICATION ACCURACY OF EACH COMPARISON ALGORITHM ON THE OBERPFAFFENHOFEN DATASET(%)

class	Super_RF	DFGCN	CVS3ANET	HybridCVNet	AMS-M2ESL	SpectralDiff	proposed
Bare ground	91.67	90.51	92.43	94.08	89.93	93.42	**95.20**
Forest	85.26	85.08	93.53	91.22	85.75	94.47	**96.66**
Buildings	59.55	87.30	95.33	89.00	78.81	95.29	**96.83**
Farmland	57.28	77.57	93.42	90.15	68.51	90.84	**94.34**
Road	49.19	66.10	87.23	77.69	60.36	84.35	**90.74**
OA	78.40	85.43	92.54	90.65	82.68	92.69	**95.14**
AA	68.59	81.31	92.39	88.43	76.67	91.67	**94.75**
KAPPA	67.24	78.66	89.23	86.24	74.37	89.38	**92.94**

TABLE IV

CLASSIFICATION ACCURACY OF EACH COMPARISON ALGORITHM ON THE SAN FRANCISCO DATASET (%)

class	Super_RF	DFGCN	CVS3ANET	HybridCVNet	AMS-M2ESL	SpectralDiff	proposed
Bare soil	77.00	65.62	**92.54**	52.19	81.99	76.06	90.33
Mountain	93.98	92.68	98.66	99.30	95.92	94.42	**99.35**
Ocean	99.01	98.01	98.84	99.58	99.36	99.09	**99.61**
Urban	98.36	95.82	99.60	99.67	96.99	98.45	**99.85**
Vegetation	69.83	68.00	95.51	92.34	81.21	90.60	**97.92**
OA	96.02	94.10	98.82	98.31	96.57	97.49	**99.42**
AA	87.64	84.03	97.03	88.62	91.09	91.72	**97.41**
KAPPA	93.71	94.09	98.15	97.33	94.63	96.04	**99.09**

TABLE V

CLASSIFICATION ACCURACY OF EACH COMPARISON ALGORITHM ON THE FLEVOLAND2 DATASET (%)

class	Super_RF	DFGCN	CVS3ANET	HybridCVNet	AMS-M2ESL	SpectralDiff	proposed
Stem beans	91.55	99.69	97.57	99.52	99.18	99.69	**99.84**
Peas	81.90	99.51	98.79	99.26	98.32	**100.00**	99.65
Forest	78.64	97.65	99.13	99.89	98.17	99.95	**99.99**
Lucerne	88.01	99.54	97.38	**99.92**	97.97	99.20	99.85
Wheat	79.14	98.06	97.65	99.51	96.34	99.94	**99.98**
Beets	82.42	98.22	95.49	**100.00**	98.92	99.20	99.93
Potatoes	79.77	97.71	98.82	97.10	96.82	**99.94**	99.60
Bare soil	74.33	97.53	**100.00**	97.86	98.05	**100.00**	99.84
Grasses	62.00	91.51	87.00	**99.84**	95.17	98.42	99.00
Rapeseed	53.65	94.24	96.18	88.54	93.99	**100.00**	99.78
Barely	85.29	99.20	98.66	92.76	98.45	**99.75**	99.44
Wheat2	55.01	91.45	97.39	99.28	94.97	**100.00**	99.93
Wheat3	84.36	99.51	99.29	99.19	98.17	99.92	**99.98**
Water	97.97	99.67	99.90	99.84	99.88	**99.97**	99.09
Buildings	86.34	80.88	37.82	91.81	91.18	82.56	**99.16**
OA	78.70	97.54	97.58	98.98	97.38	99.73	**99.74**
AA	78.69	96.29	93.40	98.35	97.04	98.57	**99.67**
KAPPA	76.73	97.32	97.36	98.88	97.15	99.71	**99.72**

detailed in this article. The learning rate for this stage is set to 0.001, and the batch size is 128. We employ the Adam optimizer throughout the entire training process. The experimental results indicate that the module typically converged within fewer than 60 epochs. Given the varying number of categories and classification difficulty across different datasets, we implement early stopping to control the training process, thereby reducing the resource waste and minimizing time consumption.

In addition, we conduct ablation experiments to achieve a more intuitive understanding of each module's function. In addition, we explore the effects of various parameters on classification accuracy and compared the running times of various methods. In the experimental analysis, we use average accuracy (AA), overall accuracy (OA), and the Kappa coefficient as evaluation metrics to assess the effectiveness of our proposed method. Furthermore, the PyTorch training framework is employed to train the model. The hardware

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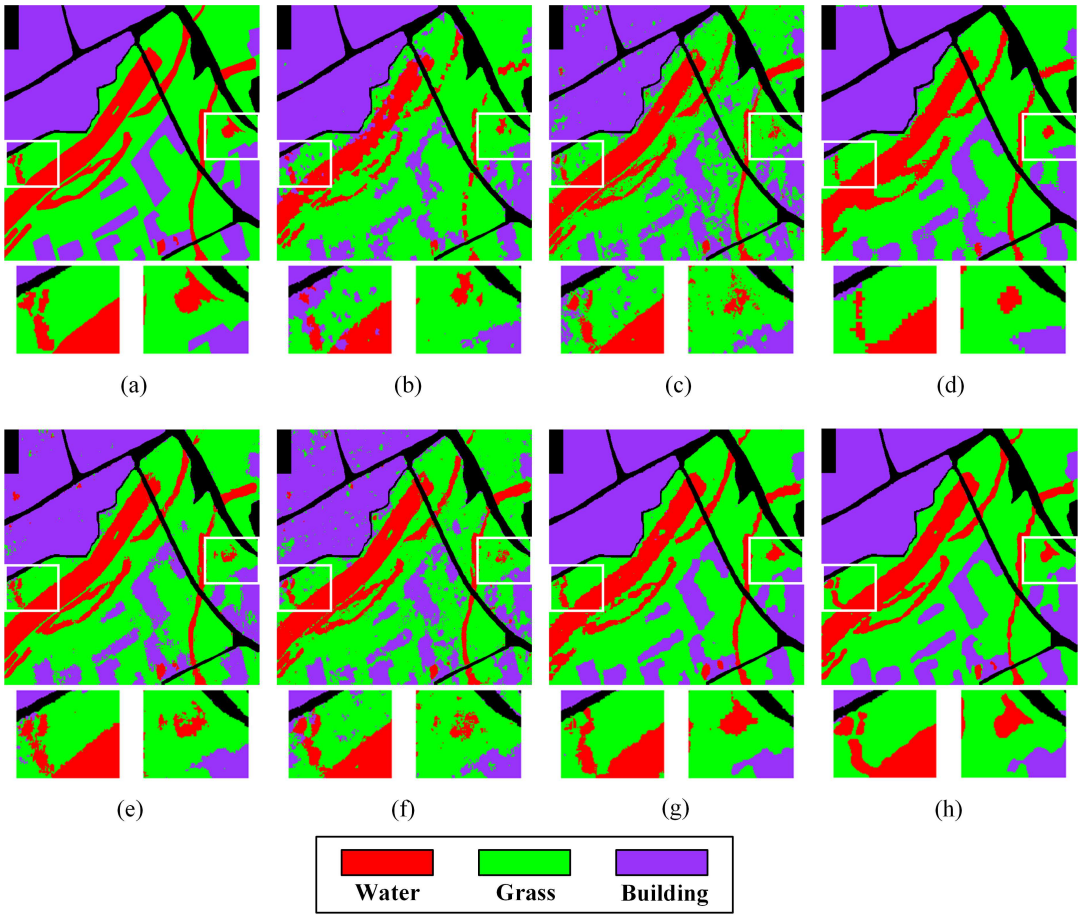


Fig. 6. Classification maps of Xi'an dataset. (a) Ground truth. (b) Super_RF. (c) DFGCN. (d) CVS3ANET. (e) HybridCVNet. (f) AMS-M2ESL. (g) Spectral Diff. (h) Proposed.

configuration comprises an Intel[1] Core[2] i7-10700 391 CPU, 64 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU featuring 12 GB of memory.

B. Experimental Result and Analysis

Tables I–V display OA, AA, Kappa, and the accuracy of each category across the five aforementioned datasets. The top results are emphasized in bold black, while the second-best outcomes are underlined. Furthermore, Figs. 6–10 visually depict comparative experiments on these datasets, alongside the classification outcomes of the proposed method. A detailed analysis of the comparative experiments for each dataset is presented below.

1) Xi'an Dataset: The classification accuracy of comparison algorithms on Xi'an dataset is summarized in Table I. The proposed method achieves the highest performance across all indicators, with SpectralDiff ranking second in most cases. Specifically, our method improves the OA value by 7.05%, 7.89%, 3.49%, 2.43%, 6.87%, and 0.78%, respectively, compared with others. Super_RF performs poorly in *water* classification due to its reliance on superpixels, which fail to capture thin structures. DFGCN, a pixel-level method, significantly enhances performance for *water*. HybridCVNet ranks second in *water* classification by leveraging 3DCNN in complex domains. AMS-M2ESL struggles to classify *grass* and *buildings* accurately, as shown in Fig. 6(f). SpectralDiff misclassifies *water* due to its lack of local receptive fields. Our method uses varied convolutional kernels and channel attention to enhance regional accuracy and reduce noise, achieving superior results, particularly in challenging *water* areas. This aligns with the experimental results in Fig. 6.

Fig. 6 illustrates the classification results of various methods on Xi'an dataset. Fig. 6(a) shows the ground truth. Fig. 6(b) presents the result of super_RF, which performs well in the building area but loses detail due to its superpixel-based approach. Fig. 6(c) depicts DFGCN, which excels in the *water* area but misclassifies *building* and *grass*. Fig. 6(d) shows CVS3ANET, which achieves noise-free results but suffers from jagged boundaries. HybridCVNet in Fig. 6(e), closely matches the ground truth in *water* boundaries but misclassifies *grass* and *buildings*. AMS-M2ESL in Fig. 6(f), introduces significant noise and boundary errors. SpectralDiff, as shown in Fig. 6(g), minimizes noise in homogeneous regions but struggles with boundaries. Finally, the proposed method in Fig. 6(h), which aligns most closely with the ground truth, achieves excellent results in both homogeneous and heterogeneous areas. This can also be seen from the boxed-out details. It

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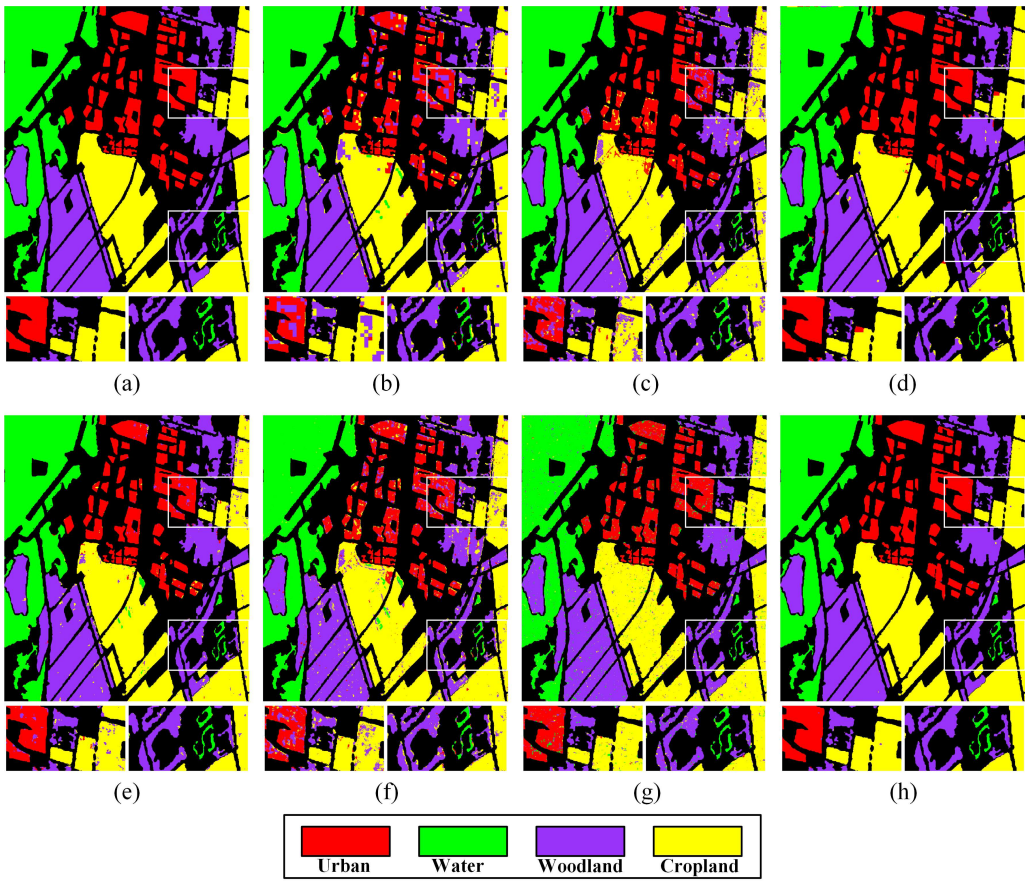


Fig. 7. Classification maps of the Flevoland1 dataset. (a) Ground truth. (b) Super_RF. (c) DFGCN. (d) CVS3ANET. (e) HybridCVNet. (f) AMS-M2ESL. (g) SpectralDiff. (h) Proposed.

enhances details and reduces noise by leveraging non-Gaussian diffusion features and scattering features, thereby achieving the superior classification accuracy.

2) Flevoland1 Dataset: The classification accuracy results on the Flevoland1 dataset are shown in Table II, where the proposed method achieves the highest accuracy on most indices, followed closely by CVS3ANET. Specifically, the proposed method outperforms others by 5.89%, 4.08%, 1.16%, 1.60%, 5.86%, and 3.18% in OA. All methods perform well, but defects remain. Super_RF and AMS-M2ESL struggle in *urban* areas due to insufficient feature learning. DFGCN ranks second in *water* classification but misclassifies *urban* areas due to its reliance on pixel-level classification. CVS3ANET and HybridCVNet exceed 98% OA, particularly in *urban* areas, where their classification capabilities have improved significantly, as they effectively utilize the scattering information of the polarized SAR data. SpectralDiff misclassifies across categories, lacking a focus on local information. The proposed method, however, captures rich data distribution and attention to local details, enhancing the classification performance.

Fig. 7(b)–(h) presents the classification results for each comparison method, with Fig. 7(a) as ground truth. Super_RF in Fig. 7(b) misclassifies *urban* areas. DFGCN in Fig. 7(c) reduces *urban* noise but introduces small particles via pixel-level classification. CVS3ANET in Fig. 7(d) classifies *urban* areas well but has confused boundaries. HybridCVNet in Fig. 7(e) generates some noise in the *cropland* area. AMS-M2ESL in Fig. 7(f) fails to accurately classify *urban* regions and misclassifies *woodlands* and *croplands*. SpectralDiff in Fig. 7(g) has small noise points due to non-Gaussian data distribution and a lack of local feature information. The classes in the two regions highlighted in Fig. 7 are notably clustered, presenting specific challenges to the classification process. It is evident that the comparison algorithms exhibit varying levels of misclassification. The proposed method in Fig. 7(h) produces almost no noise, better boundaries, and fewer misclassifications by accurately mining data characteristics and fully integrating the information from statistical features and scattering features.

3) Oberpfaffenhofen Dataset: Table III illustrates the classification accuracy of various methods on the Oberpfaffenhofen dataset in numeric form. In summary, the proposed method achieves the highest accuracy across all indexes, while the second-highest accuracy for each index is primarily attributed to CVS3ANET and SpectralDiff. When comparing the OA of different methods, the proposed method outperforms other comparison algorithms by 16.74%, 9.71%, 2.60%, 4.49%, 12.46%, and 2.45%, respectively. Super_RF performs poorly in classifying *buildings*, *farmland*, and *road*, consistent with Fig. 8(b), due to its lack of local information learning. DFGCN and AMS-M2ESL improve classification performance in these classes by enhancing local feature learn-

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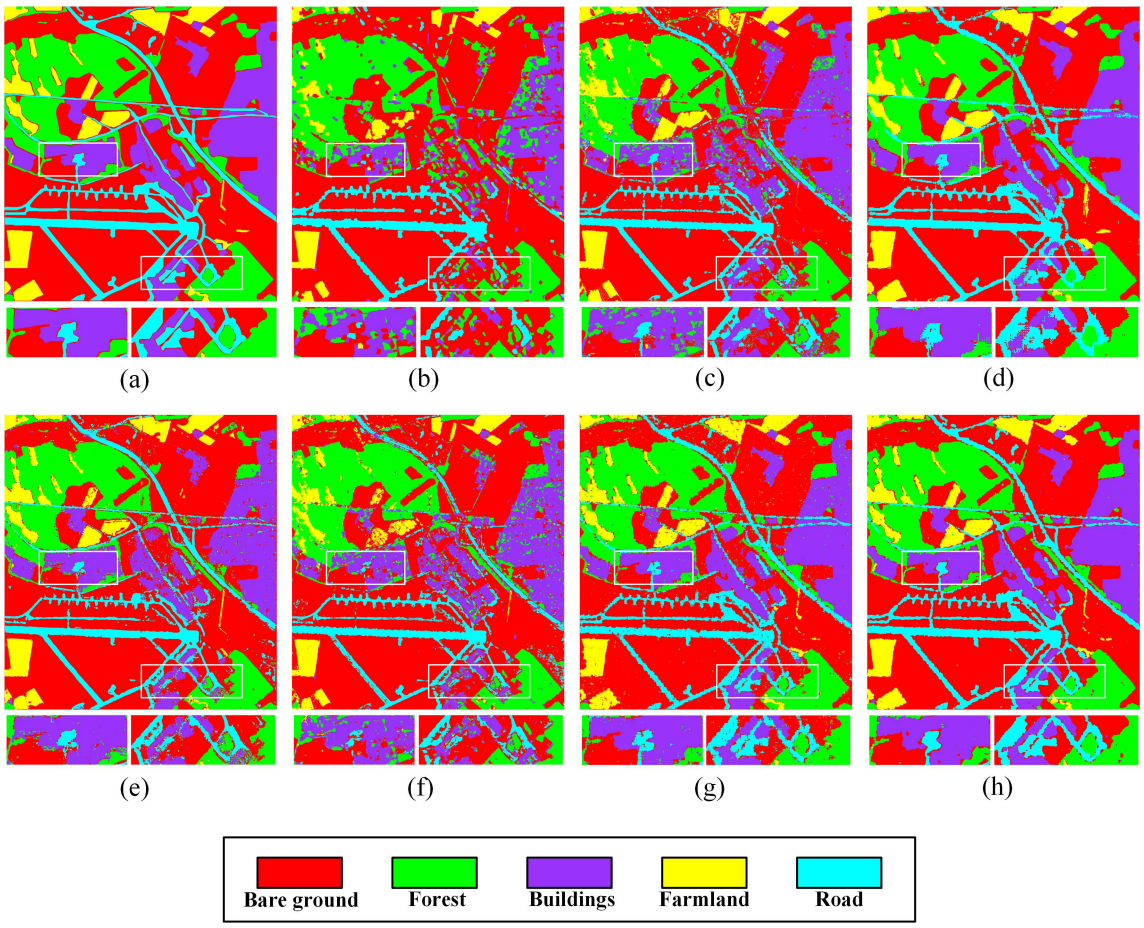


Fig. 8. Classification maps of the Oberpfaffenhofen dataset. (a) Ground truth. (b) Super_RF. (c) DFGCN. (d) CVS3ANET. (e) HybridCVNet. (f) AMS-M2ESL. (g) SpectralDiff. (h) Proposed.

ing. Both CVS3ANET and HybridCVNet employ complex value networks, learn distinguishable features, and achieve good accuracy on most indicators. SpectralDiff utilizes a Gaussian diffusion model to capture data features, achieving the second-highest OA, but underperforms in the *road* region. The proposed method significantly enhances performance on the dataset, especially in the challenging *road* region. Clearly, the effectiveness and superior performance of the proposed method are substantiated by the comparative performance of other algorithms.

Fig. 8 visually illustrates the classification performance of various comparison methods on the Oberpfaffenhofen dataset, with Fig. 8(a) representing the ground truth. Fig. 8(b) displays the classification result of Super_RF. It is evident that a significant portion of *farmland* is mistakenly classified as *forest*. In contrast to Fig. 8(a), it is observed that numerous roads appear flooded, an effect attributed to the employment of superpixels. In Fig. 8(c), DFGCN accurately classifies most instances of *forest* and *farmland* due to its pixel-level classification approach. In Fig. 8(d), the CVS3ANET demonstrates notable improvements across all categories, particularly in road classification, where its efficacy is markedly enhanced. HybridCVNet in Fig. 8(e) adopts a complex-valued network akin to CVS3ANET. While road classification improves, there are numerous misclassifications within the building areas. In Fig. 8(f), AMS-M2ESL is characterized by substantial noise within the building area and insufficient road perception. In Fig. 8(g), SepctralDiff generally performs well in classification thanks to Gaussian diffusion, though it introduces noise across different categories due to the absence of local information and the incompatibility of Gaussian processes with PolSAR data. Finally, in Fig. 8(h), the proposed method learns the more appropriate distribution for PolSAR data and integrates scattering information to enhance both the accuracy of information representation and classification. Notably, when compared with other benchmark tests, the proposed method achieves a road area classification and detail that closely aligns with ground truth, which can be clearly seen from the boxed-out regions.

4) San Francisco Dataset: Table IV presents the classification accuracy of various comparison algorithms on the San Francisco dataset. Overall, the classification accuracy in the ocean and urban categories surpasses that of other categories across all algorithms. Notably, the proposed method achieves the second-highest accuracy in the bare soil class and the best classification accuracy across the other metrics. In addition, CVS3ANET and HybridCVNet also demonstrate strong classification performance. When analyzing the OA, the proposed method outperforms the comparison algorithms by margins of 3.40%, 5.32%, 0.60%, 1.11%, 2.85%, and

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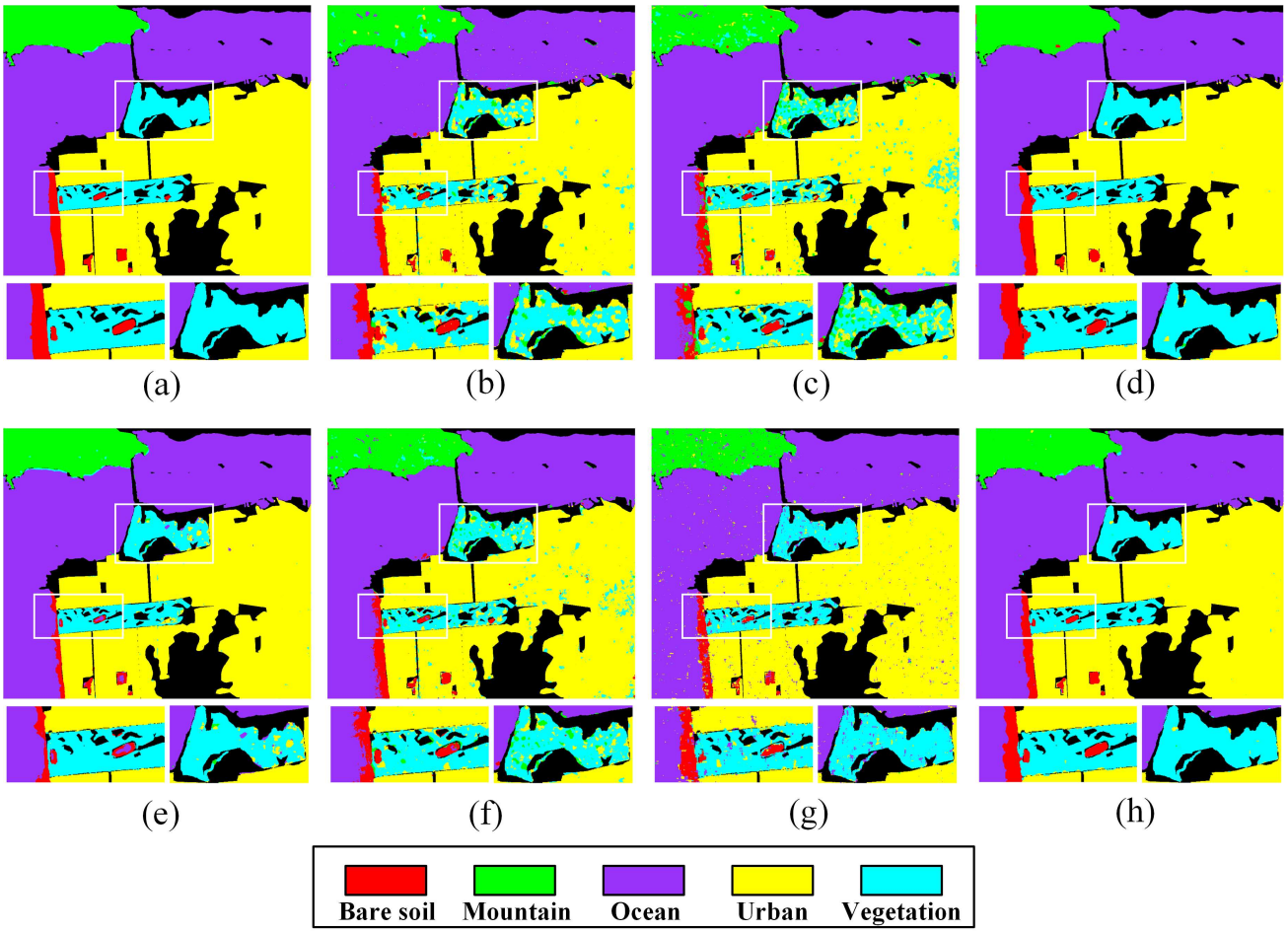


Fig. 9. Classification maps of the San Francisco dataset. (a) Ground truth. (b) Super_RF. (c) DFGCN. (d) CVS3ANET. (e) HybridCVNet. (f) AMS-M2ESL. (g) SpectralDiff. (h) Proposed.

1.93%, respectively. Specifically, Super_RF, DFGCN, and AMS-M2ESL show relatively poor classification accuracy in the *bare soil* and *vegetation* categories, as the extracted low-level features fail to adequately describe these categories. On the other hand, HybridCVNet and SpectralDiff leverage more complex networks to enhance feature representation, thereby improving performance in the *vegetation* category. The proposed method integrates scattering information and statistical distribution information, significantly boosting classification performance across all metrics.

Fig. 9(b)–(h) illustrates the classification results of different comparison algorithms on the San Francisco dataset, while Fig. 9(a) serves as the ground-truth map. Both Super_RF and DFGCN exhibit significant misclassification between *vegetation* and *urban* areas. As shown in the regions framed by white boxes, these methods also struggle to accurately classify the borders of *bare soil*. Fig. 9(d) presents the results of CVS3ANET, which generates minimal noise. However, it misclassifies some *ocean* areas as *bare soil*. HybridCVNet in Fig. 9(e), while performing well across most categories and excelling in *vegetation* area, clearly misclassifies large sections of *bare soil* as *ocean*. AMS-M2ESL achieves noise-free classification in the *ocean* category with few misclassifications, aligning with the data presented in Table IV. In contrast, SpectralDiff in Fig. 9(g), relying on a Gaussian-based diffusion model, struggles to capture the statistical characteristics of the data exactly. This limitation makes it highly susceptible to noise, resulting in widespread ground noise across various categories. Compared with SpectralDiff, the proposed method in Fig. 9(h) replaces single Gaussian noise with mixed noise, significantly enhancing classification accuracy. When focusing on the magnified areas in each figure, Fig. 9(h) consistently shows results that are closest to the ground truth.

5) Flevoland2 Dataset: Table V presents the classification accuracies of various comparative methods on the Flevoland2 dataset. As shown in the table, the proposed method and SpectralDiff achieved the best and second-best results across most classification metrics, respectively. The OA of the proposed method reached 99.74%, outperforming the other methods by 21.04%, 2.20%, 2.16%, 0.76%, 2.36%, and 0.01%, respectively. Except for Super_RF, all other methods achieved an OA above 97%. CVS3ANet, HybridCVNet, and SpectralDiff attained 100% accuracy in certain classes, while the proposed method achieved over 99% accuracy across all classes. Diffusion model-based methods, such as SpectralDiff and the proposed approach, significantly outperformed the other methods on this dataset, indicating that diffusion models can extract high-quality and highly

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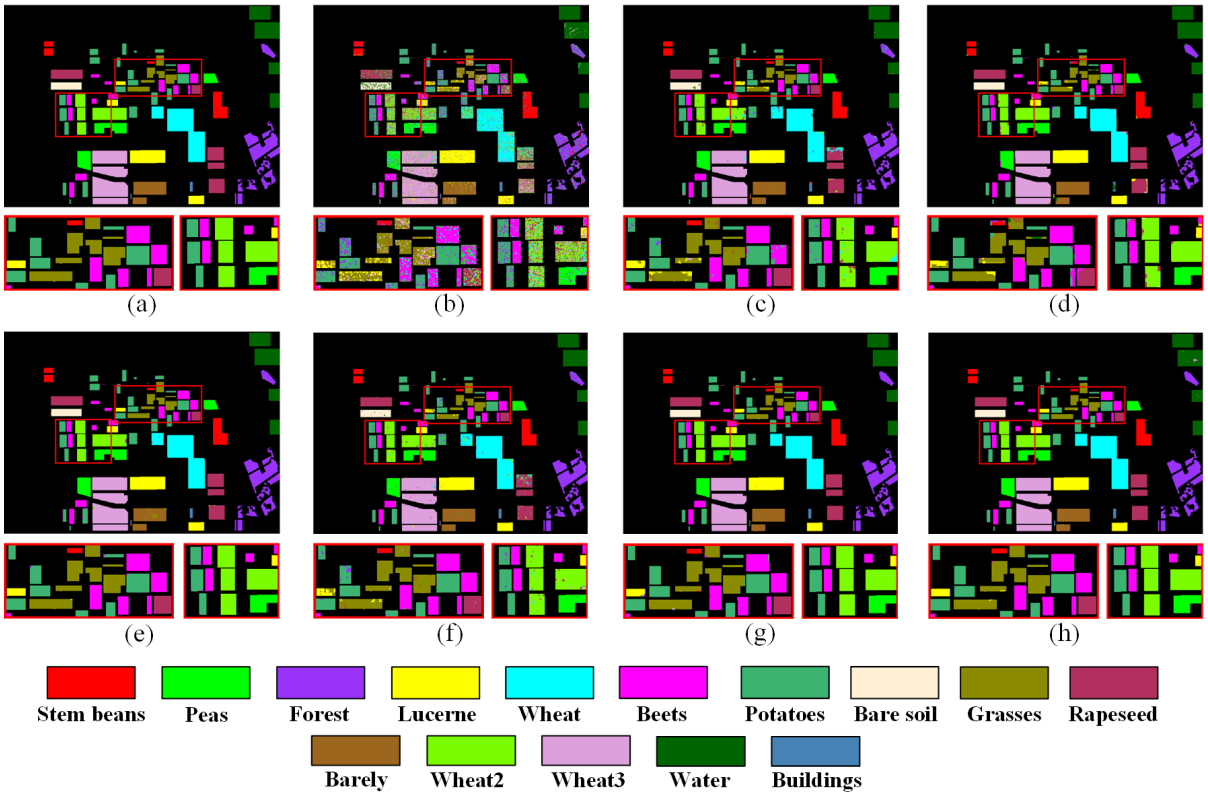


Fig. 10. Classification maps of the Flevoland2 dataset. (a) Ground truth. (b) Super_RF. (c) DFGCN. (d) CVS3ANET. (e) HybridCVNet. (f) AMS-M2ESL. (g) SpectralDiff. (h) Proposed.

TABLE VI
PERFORMANCE OF DIFFERENT INPUTS ON DIFFERENT DATASETS

Raw	GaussianDiff	NoGDiff	dual-branch	loss	Xi'an			Flevoland1			Oberpfaffenhofen			San Francisco		
					OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
✓					95.55	94.92	92.67	98.45	98.32	97.89	77.10	62.56	64.04	96.68	91.09	94.81
	✓				96.18	94.92	93.67	96.42	96.38	95.12	90.74	88.31	86.34	97.66	95.96	96.34
		✓			96.57	95.96	94.34	97.83	97.86	97.04	91.01	89.62	86.88	98.90	97.19	98.29
✓		✓	✓		96.75	**96.49**	94.65	99.70	99.69	99.59	91.51	91.65	87.79	98.95	94.61	**98.35**
✓		✓	✓	✓	**96.99**	95.98	**95.02**	**99.77**	**99.75**	**99.68**	**95.14**	**94.75**	**92.94**	**99.42**	**97.41**	97.41

discriminative feature representations—even in datasets with a large number of classes. By integrating scattering features with diffusion-based features, the proposed method provides more comprehensive representations for each class, resulting in an AA of 99.67%.

Fig. 10(b)–(h) illustrates the classification results of various comparative methods, with the ground truth in Fig. 10(a) serving as the reference. As shown in Fig. 10(b), Super_RF generates a considerable amount of noise. Upon examining the highlighted regions in each result, it is evident that DFGCN in Fig. 10(c), CVS3ANET in Fig. 10(d), and AMS-M2ESL in Fig. 10(f) suffer from notable misclassifications in multiple areas, particularly with increased confusion between *lucerne* and *grasses*. HybridCVNet extracts more discriminative features by incorporating phase information of the data, thereby mitigating such confusion. The proposed method, as shown in Fig. 10(h), combines statistical features with scattering features, enabling more accurate classification of land cover types and significantly reducing interclass confusion.

C. Ablation Analysis

1) Analysis of Diverse Modules: The ablation analysis confirms the efficacy of each module in the proposed method by conducting comparative tests on individual components. The proposed method consists of four key parts: the non-Gaussian diffusion module, raw scattering feature-based subnetwork, statistical and scattering feature joint learning module, and the loss function design. Table VI presents OA, AA, and Kappa results for various inputs across four datasets. We utilized the designed network in Section III-B as the backbone. In the table, "Raw" means that only original scattering features are used to learn the network for classification. "GaussianDiff" means a traditional Gaussian-based diffusion model for classification, which is given as a contrast to verify the effectiveness of the proposed non-Gaussian diffusion model. "NonGDiff" denotes the utilization of the proposed non-Gaussian diffusion module for classification. "dual-branch" indicates the proposed scattering and statistical joint learning network.

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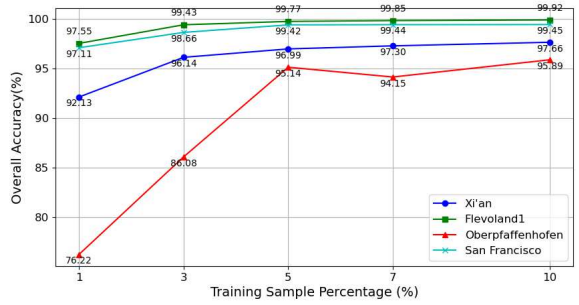


Fig. 11. Effect of training sample percentage on OA for four PolSAR datasets.

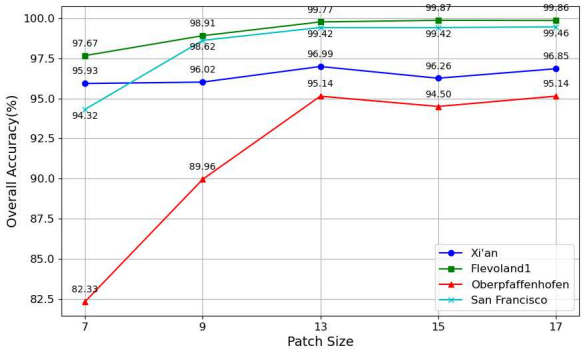


Fig. 12. Impact of patch size on OA in four PolSAR datasets.

"Loss" represents using the proposed loss function. Table VI shows that non-Gaussian diffusion features outperform Gaussian ones on Xi'an, Flevoland1, Oberpfaffenhofen, and San Francisco datasets. Compared with the fourth and fifth rows, it can be seen that incorporating intraclass and interclass losses can enhance classification performance, especially in the Oberpfaffenhofen dataset. Notably, the Oberpfaffenhofen dataset, with complex boundaries and heterogeneous regions, poses the greatest challenge. Diffusion models, particularly the non-Gaussian model, alleviate classification difficulties in complicated terrain regions and enhance accuracy, validating the proposed method's effectiveness.

2) Effect of Training Sample: Fig. 11 illustrates the variations in OA as the proportion of the training set ranges from 1% to 10% in the proposed method across four datasets. Notably, as the training set proportion increases from 1% to 5%, OA exhibits an upward trend on all datasets, with the most significant increase observed on the Oberpfaffenhofen dataset. Conversely, when the training set proportion surpasses 5%, the growth rate decelerates and stabilizes. Hence, it can be deduced that a training set ratio of 5% achieves an optimal balance, utilizing fewer training sets while maintaining an impressive OA level across all datasets.

3) Effect of Patch Size: Fig. 12 demonstrates the impact of various patch sizes selected within the network on the OA across four distinct datasets. As the patch size increases from 7×7 to 13×13, there is a notable upward trend in OA. However, when the patch size exceeds 13, OA initially declines for the Xi'an and Oberpfaffenhofen datasets before showing a subsequent increase, while it tends to stabilize for the Flevoland1 and San Francisco datasets. Consequently, setting the patch size to 13×13 optimizes the classification performance while minimizing the resource consumption.

TABLE VII

EFFECT OF DIFFERENT SHAPE PARAMETERS k OF THE GAMMA DISTRIBUTION ON THE XI'AN DATASET

	k=2	k=3	k=4	k=5	k=6
OA	96.40	96.60	**96.99**	96.52	96.37
AA	95.95	**96.32**	95.98	96.23	95.17
Kappa	94.07	94.41	**95.02**	94.28	93.99

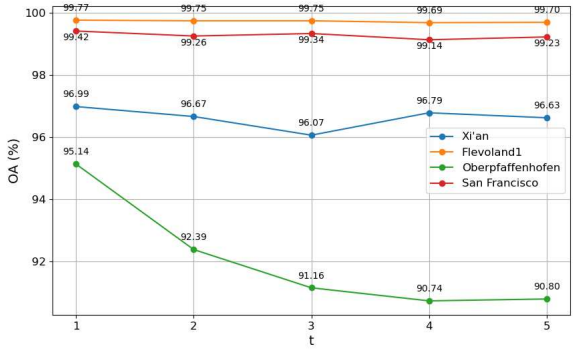


Fig. 13. Impact of different time steps on OA in four PolSAR datasets.

4) Effect of the Gamma Shape Parameter k: In this study, the scale parameter θ of the Gamma distribution is fixed at 1 to ensure unit-mean normalization, making the simulation setup more reasonable and comparable across different scenarios. The value of shape parameter k is primarily estimated using the MoMLC method, and several values in the vicinity of this estimate are further selected for analysis. Table VII illustrates the impact of the gamma distribution with varying shape parameters k on Xi'an dataset. As k increases, the OA and AA metrics generally exhibit a trend of initially rising and then slightly decreasing. The Kappa coefficient follows a similar pattern to OA and AA, but with a more pronounced decline. In the analysis of the OA, it is observed that there is minimal variation across different values of k. The OA achieves its maximum value at $k = 4$, indicating that the Gamma distribution provides a closer approximation to the true distribution of Xi'an dataset when the shape parameter k is set to 4. Also, it is consistent with the estimated $k = 4$ by the MoMLC method. Therefore, we select the shape parameter k as 4.

5) Effect of Time Steps: Fig. 13 depicts the effect of the proposed method on the OA in various datasets at different time steps t. It is evident that for the Flevoland1 and San Francisco datasets, the OA decreases only slightly with increasing time steps t, and the magnitude of this decline is minimal. For the Xi'an dataset, the OA initially drops and then rises as the time step t increases. In contrast, the OA for the Oberpfaffenhofen dataset shows a consistent downward trend with increasing time steps t, with the most significant drop

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TABLE VIII
INFERENCE TIME OF DIFFERENT ALGORITHMS ON FIVE POLSAR DATASET(S)

datasets	Super_RF	DFGCN	CVS3ANET	HybridCVNet	AMS-M2ESL	SpectralDiff	proposed
Xi'an	1.85	6.17	6.64	634.99	52.70	113.15	71.73
Flevoland1	12.96	41.36	46.77	3200.00	243.85	859.36	292.18
Oberpfaffenhofen	16.31	56.19	46.87	6456.00	408.12	1068.35	466.45
San Francisco	9.04	24.07	26.00	3080.00	184.40	452.63	245.84
Flevoland2	5.91	23.46	19.68	1170.00	36.06	375.68	47.62

occurring from $t = 1$ to $t = 2$. Overall, all datasets achieve the highest OA at $t = 1$. As the time step t increases, the degree to which images are corrupted by noise becomes more severe, thus affecting the quality of the extracted statistical features and reducing the accuracy of the classification. However, by integrating scattering features as a supplementary measure in the proposed method, the decrease in classification accuracy due to the increase in time step t is alleviated.

6) Analysis of Inference Time: Table VIII shows the inference time of different algorithms on five PolSAR datasets. Both the proposed method and spectralDiff follow a two-stage design, with the first stage dedicated to feature extraction and the second to classification, whereas the other compared methods are end-to-end. To ensure fairness and comparability, the efficiency evaluation considers only the inference time of the classification phase. Table VIII indicates that HybridCVNet has the longest inference time due to its utilization of a computationally intensive CV-VIT architecture. SpectralDiff also exhibits a higher inference time, attributed to the complexity of its classification approach. The proposed method employs a network architecture that integrates a pyramid module with an SE block. This not only reduces the inference time but also boosts the classification accuracy. The proposed method, though longer than Super_RF, DFGCN, AMS-M2ESL, and CVS3ANET algorithms, demonstrates superior classification performance across various datasets.

V. CONCLUSION

In this study, we have proposed a new scattering mechanism inspired non-Gaussian diffusion model for PolSAR image classification, which can effectively model the complex distribution of PolSAR covariance matrices. To the best of the authors' knowledge, it is the first time to develop a non-Gaussian diffusion model toward the distribution of PolSAR data. A non-Gaussian mixed distribution-based forward and reverse diffusion processes are derived to learn statistical features of PolSAR images. In addition, the proposed method further combines the non-Gaussian diffusion model with a dual-branch network to jointly learn the statistical and scattering features of PolSAR images. Experimental results on five real PolSAR datasets verify the effectiveness of the proposed method in both boundary details and heterogeneous regions.

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